

DriveBuddyAI Pav-Bhaji Text Classification Challenge

Data Analysis Report — Final Submission

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Task: Predict whether an Instagram post image is Pav-Bhaji or Not Pav-Bhaji using the available post text and metadata.

1. Executive Summary

This project implements a text-based binary classification pipeline for the DriveBuddyAI Pav-Bhaji challenge. The supplied image folders provide labels, while the Instagram JSON metadata supplies textual information associated with each image. The model does not inspect image pixels and does not use a CNN. Instead, captions and hashtags are cleaned, represented with TF-IDF word and character n-grams, and classified using conventional machine-learning algorithms.

The V6 implementation evaluates multiple text representations, n-gram configurations and model hyperparameters using stratified 5-fold cross-validation. The best configuration by cross-validation accuracy uses caption text, word n-grams (1–2), character n-grams (3–5), and balanced Logistic Regression with C=0.3. Its mean cross-validation accuracy is 72.30%. The final holdout error-analysis file contains 61 correct predictions out of 91 samples, corresponding to 67.03% accuracy.

2. Problem Definition

The target is a binary label: **1 = Pav-Bhaji** and **0 = Not Pav-Bhaji**. The challenge asks for relevant features and a trained classifier that predicts the class. Because the task is explicitly text based, visual image features are intentionally excluded.

3. Dataset and Data Preparation

Item	Value
Instagram JSON records	1,500
Labelled images	452
Pav-Bhaji images (Class 1)	183
Non-Pav-Bhaji images (Class 0)	269
Training split	80% (361 samples)
Holdout split	20% (91 samples)
Split method	Stratified, random_state=42

The V6 code matches image filenames to the filename contained in each JSON post's display URL, then joins the label with caption and tag metadata. Missing caption/tag values are converted to empty text so they can be processed consistently.

4. Features and Their Relevance

Feature	Relevance	Why it is useful
Caption / description	High	Contains semantic words and phrases describing the food or post.
Hashtags / tags	High	Cuisine and dish-specific hashtags can strongly indicate topic.
Word TF-IDF n-grams (1–2)	High	Captures individual terms and short phrases such as 'pav bhaji'.
Character TF-IDF n-grams (3–5)	Medium	Handles spelling variations, punctuation and social-media text patterns.
Caption length	Low–Medium	Can capture writing style, but is indirect evidence.
Hashtag count	Low–Medium	Provides contextual information about post composition.

Feature	Relevance	Why it is useful
Mentions / engagement	Low	May reflect account or repost patterns but can generalize poorly.
Location	Low–Medium	Can provide geographic context but may introduce bias.
Account identifiers	Excluded	Risk of leakage and account memorization.
Image pixels	Excluded	Not used because the challenge specifies a text/metadata approach.

5. Text Preprocessing

The implementation lowercases text, replaces URLs with a URL token, converts hashtag markers into a dedicated hashtag token, removes punctuation/non-alphanumeric characters except underscores, and normalizes whitespace. Caption and tag fields are cleaned independently before different representations are constructed.

Four representations are evaluated: CAPTION_ONLY, TAGS_ONLY, CAPTION_TAGS, and WEIGHTED_TEXT. The weighted representation repeats caption and tag content so that these fields receive greater influence in the textual representation.

6. TF-IDF Feature Engineering

Two TF-IDF vectorizers are used. The word-level vectorizer captures word and phrase patterns, while the character-level vectorizer uses character-within-word boundaries to capture noisy spelling and hashtag patterns. The V6 search tests word n-gram ranges (1–1), (1–2), (1–3), and (1–4), together with character ranges (3–5), (3–5), (3–6), and (3–7). Each representation/model combination is evaluated systematically.

7. Models and Model Selection

V6 evaluates Logistic Regression with multiple C values, balanced Logistic Regression with class_weight='balanced', and Linear SVM (LinearSVC) with multiple C values. Stratified 5-fold cross-validation is used on the training split. The primary selection criterion in the submitted V6 code is mean cross-validation accuracy, followed by holdout accuracy for sorting ties.

Best configuration	Value
Text representation	CAPTION_ONLY
Word n-grams	(1, 2)
Character n-grams	(3, 5)
Classifier	Balanced Logistic Regression
C	0.3
Mean 5-fold CV accuracy	72.30%
Search holdout accuracy	68.13%
Search holdout F1	65.88%

Evaluation note: The V6 search records the selected configuration with 30,000 maximum word and character features. The final model is rebuilt using the selected configuration with 50,000 maximum features. The final error-analysis CSV therefore provides the most direct record of the predictions made by the final model.

8. Final Holdout Evaluation

The final V6 error-analysis file contains 91 holdout predictions. 61 are correct and 30 are incorrect, giving a final holdout accuracy of 67.03%. The confusion matrix is:

Actual \ Predicted	Not Pav-Bhaji (0)	Pav-Bhaji (1)
Not Pav-Bhaji (0)	33	21
Pav-Bhaji (1)	9	28

Metric	Final holdout
Accuracy	67.03%
Precision (Class 1)	57.14%
Recall (Class 1)	75.68%
F1-score (Class 1)	65.12%
Correct predictions	61 / 91
False positives	21
False negatives	9

The final model identifies 28 of 37 Pav-Bhaji examples, giving a recall of 75.68%. It also predicts 21 non-Pav-Bhaji examples as Pav-Bhaji, which accounts for the lower class-1 precision of 57.14%.

9. Error Analysis

Inspection of the supplied error-analysis output shows several sources of difficulty. Some non-Pav-Bhaji posts contain the hashtag #pavbhaji or discuss Pav-Bhaji in their captions even when the image belongs to another food category. This can produce false positives. False negatives can occur when the post metadata is short, ambiguous, incomplete, or does not clearly describe the visual content.

The error analysis demonstrates an important property of this task: textual metadata is a proxy for image content. A classifier can learn strong linguistic signals while still making mistakes when captions and hashtags do not correspond precisely to the image.

10. Strengths of the Approach

- No image pixels or CNN are required, matching the challenge specification.
- Multiple text representations are compared instead of assuming that one representation is optimal.
- Character n-grams add robustness to noisy social-media text.
- Class-balanced Logistic Regression addresses the unequal class distribution during training.
- Five-fold stratified cross-validation provides a more stable model-selection signal than a single split.

11. Limitations and Future Improvements

- The labelled dataset contains only 452 images, so generalization to new Instagram data may be limited.
- The class distribution is imbalanced.
- Hashtags can be noisy or unrelated to the actual image.
- A text-only model cannot directly verify visual food content.
- A larger labelled dataset and an independent test set would provide stronger evidence of generalization.

- Future work could evaluate calibrated probabilities, additional metadata features, ensemble methods, and more robust validation such as grouped splitting by account where account information is available.

12. Conclusion

The submitted solution provides a complete end-to-end text classification workflow for the DriveBuddyAI Pav-Bhaji challenge. It identifies relevant metadata features, cleans and transforms Instagram text using TF-IDF word and character n-grams, systematically compares Logistic Regression and Linear SVM configurations, and performs holdout error analysis. The best model-selection configuration achieved 72.30% mean five-fold cross-validation accuracy. The final holdout error-analysis output records 67.03% accuracy, with 75.68% recall and 65.12% F1-score for the Pav-Bhaji class. These results show that Instagram text and hashtags contain useful signals for the task while also highlighting the inherent limitations of predicting visual content from metadata alone.

13. Reproducibility and Submission Files

The project archive contains the V6 Python source, model-results CSV, error-analysis CSV, README and .gitignore. The raw dataset is not included in the submitted GitHub project archive. The source code can be run in an environment with the required Python machine-learning libraries and access to the challenge dataset.

Prepared for: DriveBuddyAI Machine Learning Challenge

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